

Effects of baking and boiling on the nutritional and antioxidant properties of sweet potato [Ipomoea batatas (L.) Lam.] cultivars.

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The effects of baking and boiling on the nutritional and antioxidant properties of three sweet potato cultivars (Beniazuma, Koganesengan, Kotobuki) cultivated in Turkey were investigated. The samples were analyzed for proximate composition, total phenolic content, ascorbic acid, β -carotene, antiradical activity, and free sugars. The dry matter, protein, and starch contents of the sweet potatoes were significantly changed by the treatments while the ash and crude fiber contents did not differ as significantly. The β -carotene contents of baked and boiled sweet potatoes were lower than those of fresh sweet potatoes; however, the total phenolic and ascorbic acid contents of the baked and boiled sweet potatoes were higher than those of the fresh samples. Generally, the antiradical activity of the sweet potatoes increased with the treatments. Sucrose, glucose, and fructose were quantified as free sugars in all fresh sweet potatoes; however, maltose was determined in the treated samples. In terms of the analyzed parameters, there were no explicit differences among the sweet potato cultivars.